



ASIA PACIFIC UNIVERSITY OF TECHNOLOGY & INNOVATION

EE001-3-1 ANALYSIS OF CIRCUITS VERIFICATION OF TWO PORT PARAMETERS

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Objective

This experiment aims at measuring and confirming parameters of a two-port network based on experimental results. In particular, the experiment aims at identifying the z-parameters (impedance parameters) with the help of a T-network and y-parameters (admittance parameters) with the help of a p-network.

Equipment

- DC Power Supply
- Ammeter
- Voltmeter
- Breadboard
- Connecting wires
- Resistors ($R_1 = 1\text{k}\Omega$, $R_2 = 2\text{k}\Omega$, $R_3 = 3.9\text{k}\Omega$) , for the lab the resistors are ($R_1 = 0.989\text{k}\Omega$, $R_2 = 1.969\text{k}\Omega$, $R_3 = 3.86\text{k}\Omega$)

Procedure

T-Network Verification (Z-parameter)

Measurement of $I_2 = 0$

To measure the output current (I_2), an ammeter was placed in series with the output circuit. Once the correctness of the circuit connections was established the current passing through the circuit to the ground was measured. The results of (I_2) were measured and noted under each test condition.

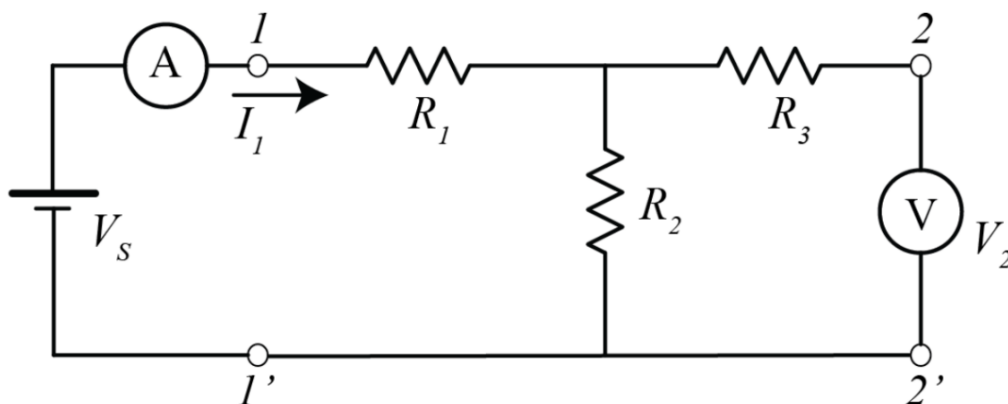


Figure 1:T-Network Verification (Z-parameter)

Measurement of $I_1 = 0$

To measure the input current (I_1), an ammeter was set in series on the input side of the circuit. Once the connections had been made, the current into the input port was measured. The results table was used to record the value of (I_1) to be analyzed later.

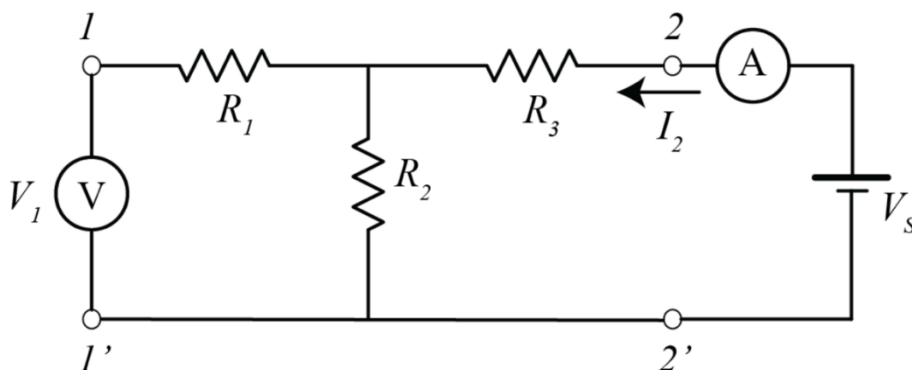
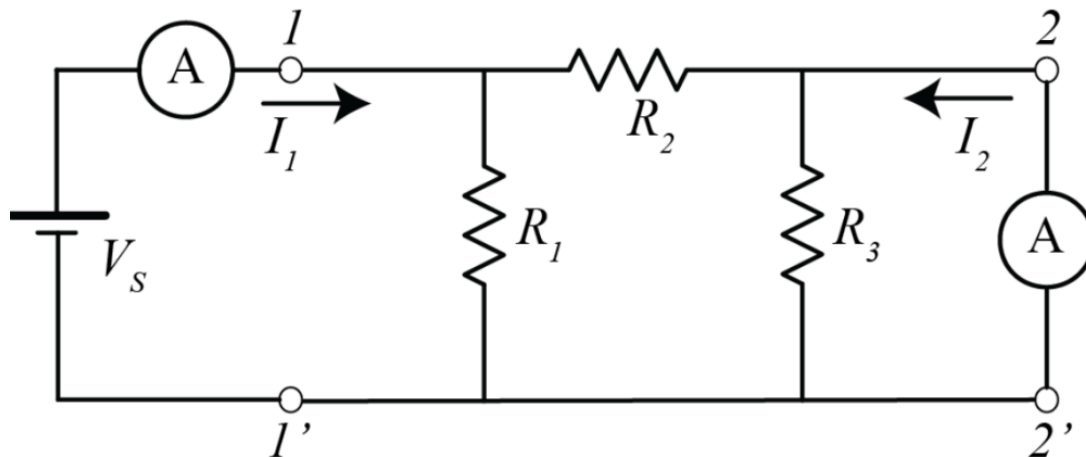


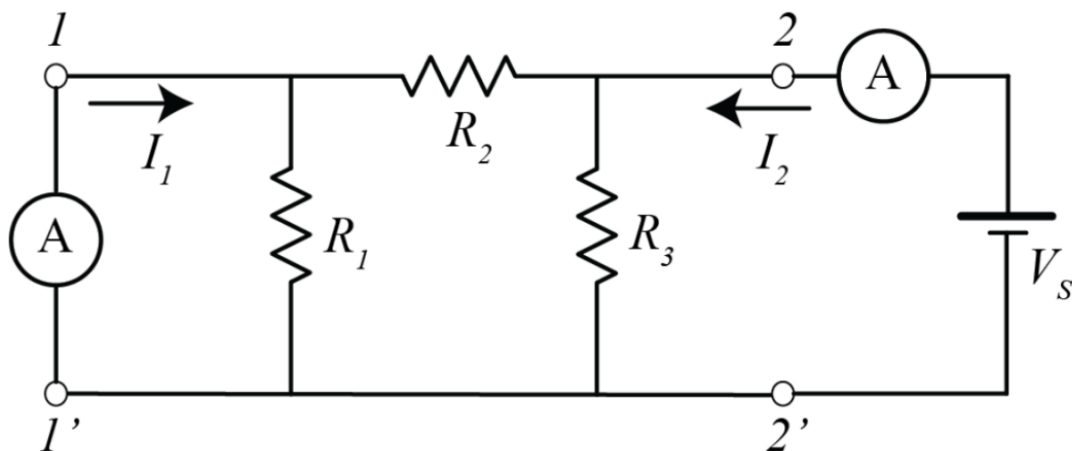
Figure 2:T-Network Verification (Z-parameter)

 π -Network Verification (Y-parameter)**Measurement of $V_1 = 0$**

The circuit was initially built on a breadboard based on the two-port network diagram given. The input port of the circuit was connected to a DC power supply. The voltmeter was then attached across the input terminals to measure the input voltage (V_1). The results were observed and the reading of (V_1) was noted and tabulated in the results table.

Figure 3: π -Network Verification (Y-parameter)**Measurement of $V_2 = 0$**

The output voltage (V_2) was determined by attaching voltmeter to the terminals of the circuit across the output. The voltage at the output port was then monitored and noted on the circuit after it was powered. The resultant value of (V_2) was then keyed in the results table.

Figure 4: π -Network Verification (Y-parameter)**Voltage Variation**

Measurements of (V_1), (I_1), (V_2) and (I_2) were repeated whereby the DC power supply was adjusted to various voltage levels. There was a supply voltage of 5 V, 10 V and 15 V which was

adjusted, and the readings respectively were noted. This experiment made sure that the action of the two port network could be monitored when the input voltage varied.

Results

LAB Results

Table 1: T-Network

$I_2 = 0$	V1	5V	10V	15V
	I 1	1.71	3.40	5.10
	V2	3.38	6.69	9.97
$I_1 = 0$	V 1	5V	10V	15V
	I 2	0.86	1.71	2.57
	V1	1.72	3.41	5.07

Figure 6: T-Network, $I_2 = 0$

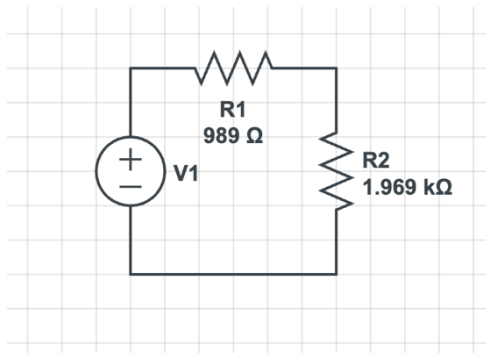


Figure 5: T-Network, $I_1 = 0$

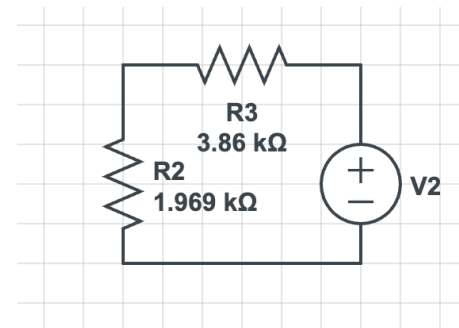
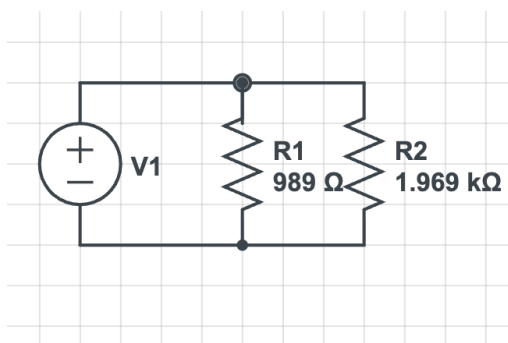
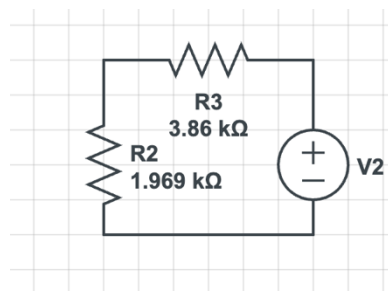


Table 2: π -Network Verification

$V_2 = 0$	V2	5V	10V	15V
	I1	-2.57	-5.11	-7.73
	I2	3.85	7.70	11.55
$V_1 = 0$	V1	5V	10V	15V
	I1	7.21	14.86	22.47
	I2	-2.27	-5.09	-7.53

Figure 8: Y-Network, $V_2 = 0$ Figure 7: Y-Network, $V_1 = 0$ 

Pre-Lab Results

$I_2 = 0$	V1	5V	10V	15V
	I 1	1.62	3.33	5
	V2	3.32	6.67	10
$I_1 = 0$	V 2	5V	10V	15V
	I 2	0.85	1.70	2.54
	V1	1.69	3.39	5.08

Table 3 :T-Network

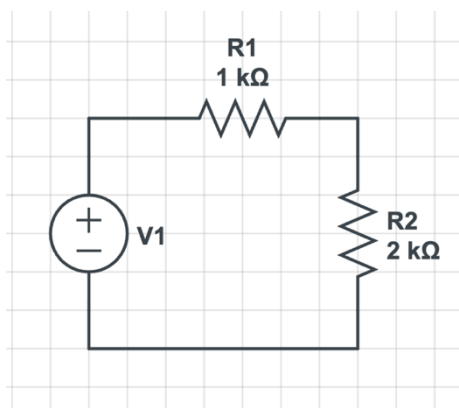
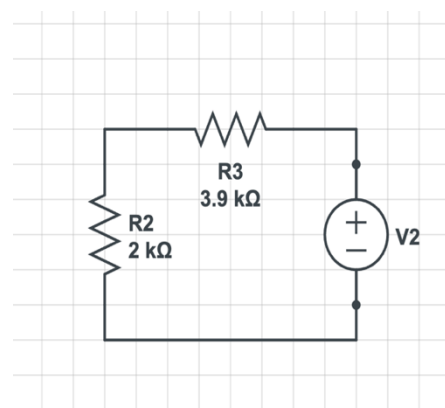
Figure 10: T-Network, $I_2 = 0$ Figure 9: T-Network, $I_1 = 0$ 

Table 4: π -Network Verification

V2 = 0	V2	5V	10V	15V
	I1	-2.50	-5	-7.5
	I2	3.78	7.56	11.35
V1 = 0	V1	5V	10V	15V
	I1	7.5	15	22.5
	I2	-2.50	-5	-7.5

Figure 12 Y-Network, V2 = 0

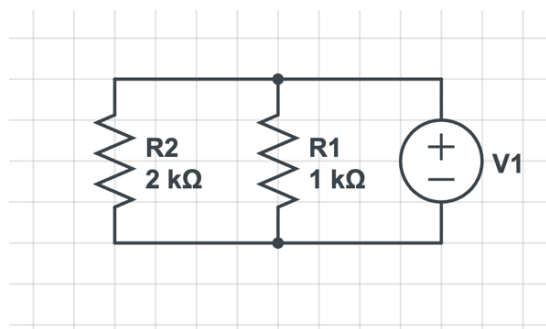
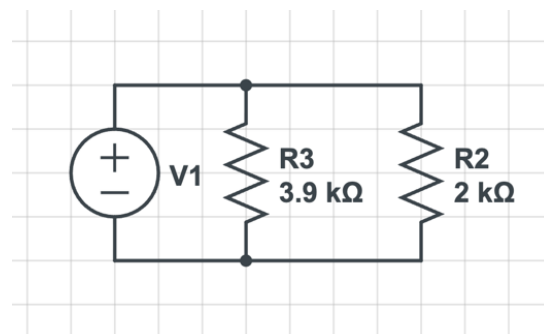


Figure 11 Y-Network, V1 = 0



Pre-Lab Z-Parameter and Y-parameter Matrix

The Z-parameters were computed with respect to theoretical values of the resistors (pre-lab) and experimental values in the laboratory session. The obtained matrices of Z-parameters are presented below.

	VS		
	5V	10V	15V
Z11	$5/1.62=3.09 \text{ k}\Omega$	$10/3.33=3 \text{ k}\Omega$	$15/5=3 \text{ k}\Omega$
Z12	$1.69/0.85=1.99 \text{ k}\Omega$	$3.39/1.70=1.99 \text{ k}\Omega$	$5.08/2.54=2 \text{ k}\Omega$
Z21	$3.32/1.62=2.05 \text{ k}\Omega$	$6.67/3.33=2 \text{ k}\Omega$	$10/5=2 \text{ k}\Omega$
Z22	$5/0.85=5.88 \text{ k}\Omega$	$10/1.7=5.88 \text{ k}\Omega$	$15/2.54=5.91 \text{ k}\Omega$

Y11	$7.5/5=1.5\text{mS}$	$15/10=1.5\text{ mS}$	$22.5/15=1.5\text{ mS}$
Y12	$-2.5/5=-0.5\text{ mS}$	$-5/10=-0.5\text{ mS}$	$-7.5/15=-0.5\text{ mS}$
Y22	$-2.5/5=-0.5\text{ mS}$	$-5/10=-0.5\text{ mS}$	$-7.5/15=-0.5\text{ mS}$
Y21	$3.78/5=0.76\text{ mS}$	$7.56/10=0.76\text{ mS}$	$11.35/10=0.76\text{ mS}$

Table 5: Pre-lab Z-parameter and Y-parameter Matrix

Lab Z-Parameter and Y-parameter Matrix

	VS		
	5V	10V	15V
Z11	$5/1.71=2.92\text{ k}\Omega$	$10/3.40=2.94\text{ k}\Omega$	$15/5.10=2.94\text{ k}\Omega$
Z12	$1.72/0.86=2\text{ k}\Omega$	$3.41/1.71=1.99\text{ k}\Omega$	$5.07/2.57=1.97\text{ k}\Omega$
Z21	$3.38/1.71=1.98\text{ k}\Omega$	$6.69/3.40=1.97\text{ k}\Omega$	$9.97/5.10=1.95\text{ k}\Omega$
Z22	$5/0.86=5.81\text{ k}\Omega$	$10/1.71=5.85\text{ k}\Omega$	$15/2.57=5.84\text{ k}\Omega$
Y11	$7.21/5=1.44\text{ mS}$	$14.86/10=1.49\text{ mS}$	$22.47/15=1.50\text{ mS}$
Y12	$-2.57/5=-0.51\text{ mS}$	$-5.11/10=-0.51\text{ mS}$	$-7.73/15=-0.52\text{ mS}$
Y22	$-2.27/5=-0.45\text{ mS}$	$-5.09/10=-0.51\text{ mS}$	$-7.53/15=-0.50\text{ mS}$
Y21	$3.85/5=0.77\text{ mS}$	$7.70/10=0.77\text{ mS}$	$11.55/15=0.77\text{ mS}$

Table 6:lab Z-parameter and Y-parameter Matrix

Discussion

In the experiment, two-port networks were studied in terms of the T-network and p-network models. Measurements of voltages and currents in the T-network determined the z-parameters, and measurements in the p-network determined the y-parameters. Based on the findings, one can note that the dependence of voltage on current is linear and it proves the Ohm Law. The current and the output voltage rose in proportion to the input voltage as the input voltage rose between 5V and 15V. This means that the circuit is a linear network. The z-parameters that were obtained were relatively stable across the various levels of voltage, which confirms that z-parameters are not dependent on the magnitude of the input source. In the same vein, the y-parameters were consistent and this tested the theoretical model.

Small differences in measurements are possible because of experimental errors (tolerance of resistor, error in measurement, connection problems) or problems associated with setup on the breadboard. These minor deviations notwithstanding, the total results are quite close to the theoretical assumptions. The given experiment therefore manages to illustrate the theoretical concept of two-port network parameters and confirm the theoretical calculation with practical data.

Conclusion

To sum up, the experiment was able to confirm parameters of a two-port network in two T-network and P-network networks. The voltage and current were measured and the z-parameters and y-parameters calculated. The findings indicated that there was a linear relationship between voltage and current and the circuit acted as expected. The parameters calculated were stable with varying input voltages, which means that they do not depend on the supply voltage. As much as some minor errors were noted owing to practical constraints, the experimental findings were very close to the theoretical expectations. This proves that two-port network analysis is reliable in circuit design. Overall, the aim of the experiment was met, and the improved understanding of the parameters of impedance and admittance was obtained.

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